

Recommendations — tnbc-brca1-oligomet-liver-r7p3

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

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Profile snapshot

- **Primary site:** breast
- **Histology:** invasive ductal carcinoma, no special type, triple-negative; basal-like by molecular subtyping
- **Stage:** cT2-3 cN1 cM1 — newly diagnosed; right breast primary with ipsilateral axillary nodal involvement and a solitary 1.5cm hepatic lesion consistent with oligometastatic de novo M1 disease (liver-only single-site)
- **ECOG:** 1
- **Age band:** 40-49
- **Sex:** female
- **Biomarkers:** BRCA1 mutated; germline vs somatic not specified; TP53 mutated; PIK3CA mutated (ngs_pending); MYC amplification copy-number amplified; IRS2 amplification copy-number amplified; PTEN copy-number loss; Intrinsic molecular subtype basal-like; TMB 14 mut/Mb; MSI status MSS; Stromal TILs 3+ stromal TILs (TILs-WG criteria, high); ER negative; PR negative; HER2 negative
- **Efficacy/toxicity weight:** 0.7

9 ranked therapeutic options across 4 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

Tmb High interventions

1. Pembrolizumab + carboplatin/gemcitabine (KEYNOTE-355 backbone) — gated on 22C3 CPS ≥ 10

Likelihood of effect: High in the CPS ≥ 10 subset: OS HR 0.73 / mOS 23.0 vs 16.1 mo (pmid:35857659); zero labeled benefit if CPS returns < 10 , and CPS is pending.

Toxicity burden: High (neutropenia ~41%, immune-mediated events, treatment-related death 0.4%)

Counter-productive MoA:Low (In an MSS tumor a B2M/HLA class I-loss subclone would blunt the checkpoint arm, but no persona vetoed on mechanism.)

Overall:The maturest 1L survival evidence in mTNBC and the board's only unanimous endorsement, though the entire labeled benefit is gated on a pending CPS ≥ 10 result.****

Key references: PMID 33278935 · PMID 35857659 · PMID 29713086 · NCT02819518

Evidence in detail

Pembrolizumab + carboplatin/gemcitabine (KEYNOTE-355 backbone) — gated on 22C3 CPS ≥ 10

****Cortes J et al. *Lancet* 2020****

- **Disease context:**Previously untreated locally recurrent inoperable or metastatic TNBC (1L) — 1L

mTNBC; PD-L1 stratified by 22C3 CPS; ECOG 0-1; primary endpoint in CPS ≥ 10 subgroup.

- **n:** 847
- **Toxicities:** any treatment-related AE (grade ≥ 3): 68.1% (n=562); neutropenia (grade ≥ 3): 41.0% (n=562); immune-mediated AE (grade ≥ 3): 5.3% (n=562); treatment-related deaths (grade 5): 0.4% (n=562)
- **Efficacy:** HR_PFS 0.65 (95% CI 0.49-0.86); PFS 9.7 months

****Cortes J et al. *N Engl J Med* 2022****

- **Disease context:** Previously untreated mTNBC, PD-L1 CPS ≥ 10 final OS (1L) — Same 847-patient cohort; final OS analysis at 44-mo median follow-up.
- **n:** 847
- **Toxicities:** any treatment-related AE (grade ≥ 3): 68.1% (n=562); treatment-related deaths (grade 5): 0.4% (n=562)
- **Efficacy:** HR_OS 0.73 (95% CI 0.55-0.95); OS 23.0 months

****Tutt A et al. *Nat Med* 2018****

- **Disease context:** Metastatic or recurrent locally advanced TNBC or BRCA1/2-mutant breast cancer (1L-2L) — Unselected advanced TNBC (376 pts); 43 patients with germline BRCA1/2 mutations identified by prospective biomarker analysis.
- **n:** 376
- **Toxicities:** neutropenia (grade ≥ 3): 12.0% (n=188); thrombocytopenia (grade ≥ 3): 7.0% (n=188); neuropathy (grade ≥ 3): 11.0% (n=188)
- **Efficacy:** ORR 31.4%; ORR 68%; ORR 33%; HR_PFS 0.44

Brca1 Mutation interventions

1. Olaparib 300 mg PO BID (OlympiAD) — gated on germline BRCA1 confirmation

Likelihood of effect: High in gBRCA1 disease: PFS HR 0.58, ORR 60% (pmid:28578601), replicated by EMBRACA; off-label and weaker-supported if BRCA1 is somatic-only.

Toxicity burden: Low (anemia 16.1%; nausea/fatigue mostly low-grade)

Counter-productive MoA: Low (No mechanism blunts the therapeutic goal; later BRCA1 reversion is a resistance route, not a counter-productive vector.)

Overall: ****Category-1, replicated, oral single-agent efficacy directly on the BRCA1 axis with the gentlest hematologic profile of the PARP options — conditional on the germline call.****

Key references: PMID 28578601 · PMID 34081848 · NCT02000622

2. Talazoparib 1 mg PO daily (EMBRACA) — gated on germline BRCA1 confirmation

Likelihood of effect: High in gBRCA1 disease: PFS HR 0.54, ORR 62.6% (pmid:30110579) — deepest single-agent response; OS gain not significant.

Toxicity burden: Moderate (anemia ~39%, neutropenia ~21%, thrombocytopenia ~15%)

Counter-productive MoA:Low (No mechanism-level counter-productive vector; the risktaker/critic split was on tolerability and ORR depth, not mechanism.)

Overall:Equal-labeled PARP alternative with the deepest ORR on the BRCA1 axis; ranks below olaparib only on a heavier hematologic burden, not on evidence.****

Key references: PMID 30110579 · NCT01945775 · NCT03990896

3. Carboplatin AUC2 + gemcitabine d1,8 q21d (PD-L1-agnostic, germline-agnostic BRCA1 backbone)

Likelihood of effect: Moderate-to-high in BRCA1-mut disease: ORR 68% vs 33% for carboplatin over taxane (pmid:29713086), from an n=43 subgroup with a wide CI.

Toxicity burden: Moderate (neutropenia 30-40%, thrombocytopenia ~15%)

Counter-productive MoA:Low (No counter-productive mechanism; the tradeoff is forgone OS benefit versus the CPS-gated ICI arm, not a mechanistic vector.)

Overall:The biomarker-agnostic BRCA1 platinum backbone that no pending test can foreclose; a floor and an induction platform rather than the efficacy ceiling.****

Key references: PMID 29713086 · PMID 35093516

4. Niraparib + pembrolizumab (TOPACIO) — investigational/off-label, later-line

Likelihood of effect: Moderate but uncertain: BRCA-mut subgroup ORR 47% (n=15) vs published cohort 21% (n=42, pmid:31194225); single-arm, no comparator.

Toxicity burden: Moderate (anemia, thrombocytopenia, immune-related AEs stacked)

Counter-productive MoA:Moderate (In an MSS tumor a B2M/HLA-loss subclone would blunt the checkpoint half; critic dissented on the thin single-arm evidence base.)

Overall:A mechanistically clean chemo-free PARP+IO bet for this exact BRCA1/TMB/TIL phenotype, but the 47% is an n=15 subgroup and niraparib is off-label — a later-line maintenance question, not a 1L move.**

Key references: PMID 31194225 · PMID 32771088

5. PARP1-selective inhibitor trial — HS-10502 on NCT05740956 (saruparib/PETRA as class precedent), later-line

Likelihood of effect: Uncertain: class-precedent ORR 48.4% (saruparib PETRA, abstract, mixed HR+/TNBC); HS-10502's own activity unpublished.

Toxicity burden: Not characterized (per-term G3+ table not released; mechanism predicts less anemia)

Counter-productive MoA:Moderate (Critic and conservative dissented on the unproven toxicity-sparing claim; no

per-term safety data to confirm PARP1-selectivity spares marrow.)

Overall:The enrollable toxicity-sparing PARP bet for a young patient facing years on the DDR axis, but the efficacy is cross-subtype abstract-level and the enrollable agent has no readout of its own.****

Key references: NCT05740956 · PMID 28578601

6. Olaparib + ceralasertib (ATR-i) on the recruiting germline-BRCA arm (NCT04090567) — later-line

Likelihood of effect: Low-to-moderate: TNBC ORR 17.1% (pmid:37773077), single-arm 2L+; the randomized VIOLETTE test did not beat olaparib alone.

Toxicity burden: Moderate (anemia and hypertension added over single-agent olaparib)

Counter-productive MoA: Moderate (Conservative flagged that the added ATR toxicity buys efficacy the randomized trial could not demonstrate over PARP monotherapy.)

Overall:The best-evidenced DDR-combination entry and a recruiting home for the BRCA1+TP53 ATR rationale, but VIOLETTE was negative over olaparib alone and it is 2L+ only.****

Key references: PMID 37773077 · NCT04090567 · NCT03330847

Evidence in detail

Olaparib 300 mg PO BID (OlympiAD) — gated on germline BRCA1 confirmation

****Robson M et al. *N Engl J Med* 2017****

- **Disease context:**HER2-negative metastatic breast cancer with germline BRCA1/2 mutation (TNBC or HR+; up to 2 prior chemo lines) (1L-3L) — Germline BRCA1/2 carriers; 50% TNBC, 50% HR+; ECOG 0-1; prior anthracycline and taxane required; up to 2 prior chemotherapy lines for metastatic disease.
- **n:** 302
- **Toxicities:**any treatment-related AE (grade ≥3): 36.6% (n=205); **anemia** (grade ≥3): 16.1% (n=205); **neutropenia** (grade ≥3): 9.3% (n=205); **fatigue** (grade any): 28.8% (n=205); **nausea** (grade any): 58.0% (n=205); **treatment discontinuation due to AE** (grade any): 4.9% (n=205)
- **Efficacy:**HR_PFS 0.58 (95% CI 0.43-0.8); **PFS** 7.0 months; **ORR** 59.9%

****Tutt ANJ et al. *N Engl J Med* 2021****

- **Disease context:**High-risk early-stage HER2-negative breast cancer with germline BRCA1/2 mutation (adjuvant) (adj) — Germline BRCA1/2 carriers; high-risk early breast cancer (TNBC or HR+); residual disease after neoadjuvant chemo OR pT2/pN1 after adjuvant chemo.
- **n:** 1836
- **Toxicities:**any treatment-related AE (grade ≥3): 24.5% (n=911); **anemia** (grade ≥3): 8.7% (n=911); **neutropenia** (grade ≥3): 4.8% (n=911); **treatment discontinuation due to AE** (grade any): 9.9% (n=911); **MDS/AML** (grade any): 0.2% (n=911)
- **Efficacy:**HR_DFS 0.58 (95% CI 0.46-0.74); **DFS** 3-yr iDFS 85.9%; **HR_OS** 0.68 (95% CI 0.47-0.97)

Talazoparib 1 mg PO daily (EMBRACA) — gated on germline BRCA1 confirmation

Litton JK et al. *N Engl J Med* 2018

- **Disease context:** Locally advanced or metastatic HER2-negative breast cancer with germline BRCA1/2 mutation (1L-3L) — Germline BRCA1/2 carriers; 45% TNBC, 55% HR+; ECOG 0-2; prior taxane or anthracycline required unless contraindicated; up to 3 prior chemo lines.
- **n:** 431
- **Toxicities:** anemia (grade ≥ 3): 39.0% (n=286); neutropenia (grade ≥ 3): 21.0% (n=286); thrombocytopenia (grade ≥ 3): 15.0% (n=286); fatigue (grade any): 50.3% (n=286); treatment discontinuation due to AE (grade any): 7.7% (n=286)
- **Efficacy:** HR_PFS 0.54 (95% CI 0.41-0.71); PFS 8.6 months; ORR 62.6%; HR_OS 0.76 (95% CI 0.55-1.06)

Carboplatin AUC2 + gemcitabine d1,8 q21d (PD-L1-agnostic, germline-agnostic BRCA1 backbone)

Tutt A et al. *Nat Med* 2018

- **Disease context:** Metastatic or recurrent locally advanced TNBC or BRCA1/2-mutant breast cancer (1L-2L) — Unselected advanced TNBC (376 pts); 43 patients with germline BRCA1/2 mutations identified by prospective biomarker analysis.
- **n:** 376
- **Toxicities:** neutropenia (grade ≥ 3): 12.0% (n=188); thrombocytopenia (grade ≥ 3): 7.0% (n=188); neuropathy (grade ≥ 3): 11.0% (n=188)
- **Efficacy:** ORR 31.4%; ORR 68%; ORR 33%; HR_PFS 0.44

Geyer CE Jr et al. *Ann Oncol* 2022

- **Disease context:** Stage II-III triple-negative breast cancer, neoadjuvant (neoadj) — Same 634-patient cohort, 4.5-yr median follow-up
- **n:** 634
- **Toxicities:** second primary malignancy (incl MDS/AML) (grade any): 1.6% (n=628)
- **Efficacy:** HR_EFS 0.63 (95% CI 0.43-0.92); HR_EFS 0.57 (95% CI 0.36-0.91)

Niraparib + pembrolizumab (TOPACIO) — investigational/off-label, later-line

Vinayak S et al. *JAMA Oncol* 2019

- **Disease context:** Advanced or metastatic triple-negative breast cancer (2L+) — Advanced TNBC; BRCA-mut (n=15) and BRCA-WT (n=27) both eligible; up to 2 prior cytotoxic lines.
- **n:** 55
- **Toxicities:** anemia (grade ≥ 3): 18.0% (n=55); thrombocytopenia (grade ≥ 3): 15.0% (n=55); fatigue (grade ≥ 3): 7.0% (n=55); immune-related AE (any) (grade any): 15.0% (n=55); immune-related AE (grade ≥ 3): 4.0% (n=55)
- **Efficacy:** ORR 21%; ORR 47%; ORR 11%

Domchek SM et al. *Lancet Oncol* 2020

- **Disease context:** Germline BRCA-mutated HER2-negative metastatic breast cancer (1L-3L) — Germline BRCA1/2 carriers; HER2-neg mBC; up to 2 prior chemo lines; ECOG 0-1.
- **n:** 34
- **Toxicities:** any treatment-related AE (grade ≥ 3): 32.0% (n=34); anemia (grade ≥ 3): 12.0% (n=34);

neutropenia (grade ≥ 3): 9.0% (n=34); **pancreatitis** (grade ≥ 3): 6.0% (n=34); **treatment discontinuation due to AE** (grade any): (n=34)

- **Efficacy:**DCR 80% (95% CI 64.3-90.9); **ORR** 63%; **PFS** 8.2 months

PARP1-selective inhibitor trial — HS-10502 on NCT05740956 (saruparib/PETRA as class precedent), later-line

****Robson M et al. *N Engl J Med* 2017****

- **Disease context:**HER2-negative metastatic breast cancer with germline BRCA1/2 mutation (TNBC or HR+; up to 2 prior chemo lines) (1L-3L) — Germline BRCA1/2 carriers; 50% TNBC, 50% HR+; ECOG 0-1; prior anthracycline and taxane required; up to 2 prior chemotherapy lines for metastatic disease.
- **n:** 302
- **Toxicities:**any treatment-related AE (grade ≥ 3): 36.6% (n=205); **anemia** (grade ≥ 3): 16.1% (n=205); **neutropenia** (grade ≥ 3): 9.3% (n=205); **fatigue** (grade any): 28.8% (n=205); **nausea** (grade any): 58.0% (n=205); **treatment discontinuation due to AE** (grade any): 4.9% (n=205)
- **Efficacy:**HR_PFS 0.58 (95% CI 0.43-0.8); **PFS** 7.0 months; **ORR** 59.9%

Olaparib + ceralasertib (ATR-i) on the recruiting germline-BRCA arm (NCT04090567) — later-line

****Ring et al. *Clinical Cancer Research* 2023****

- **Disease context:**Triple-negative advanced breast cancer (2L+) — plasmaMATCH Cohort E; 70 evaluable mTNBC patients, mostly HRR-wildtype (73% no germline/somatic BRCA), 14% germline BRCA1/2, 4% somatic BRCA2. 27% relapsed within 12 mo of adjuvant therapy; 56% one prior chemo line.
- **n:** 70
- **Toxicities:**hypertension (grade ≥ 3): 14% (n=70); **anemia** (grade ≥ 3): 13% (n=70)
- **Efficacy:**ORR 17.1% (95% CI 10.4-25.5); **PFS** 4.3 months; **DoR** 9.1 months

Basal Like interventions

1. Datopotamab deruxtecan +/- durvalumab on TROPION-Breast05 (NCT06103864) — 1L on-protocol route if CPS ≥ 10

Likelihood of effect: Unquantified: 1L pivotal readout pending (nct:NCT06103864); the control arm reproduces the KEYNOTE-355 floor, the experimental-arm benefit is untested.

Toxicity burden: Moderate (stomatitis, ocular events, ILD/pneumonitis; fewer hematologic events than chemo)

Counter-productive MoA:Moderate (TROP2-ADC has no DDR mechanism, so the experimental arm may spend the curative-intent window off the BRCA1 axis (risktaker dissent).)

Overall:**The recruiting 1L on-protocol route whose control arm reproduces standard of care, honoring the trials-first preference; the experimental arm's TROP2 mechanism sits off the BRCA1 axis and is untested.**

Key references: NCT06103864 · PMID 40297626 · PMID 33278935

Evidence in detail

Datopotamab deruxtecan +/- durvalumab on TROPION-Breast05 (NCT06103864) — 1L on-protocol route if CPS ≥ 10

Cortes J et al. *Lancet* 2020

- **Disease context:** Previously untreated locally recurrent inoperable or metastatic TNBC (1L) — 1L mTNBC; PD-L1 stratified by 22C3 CPS; ECOG 0-1; primary endpoint in CPS ≥ 10 subgroup.
- **n:** 847
- **Toxicities:** any treatment-related AE (grade ≥ 3): 68.1% (n=562); neutropenia (grade ≥ 3): 41.0% (n=562); immune-mediated AE (grade ≥ 3): 5.3% (n=562); treatment-related deaths (grade 5): 0.4% (n=562)
- **Efficacy:** HR_PFS 0.65 (95% CI 0.49-0.86); PFS 9.7 months

Oligometastatic Liver interventions

1. SBRT to the solitary liver lesion as consolidation on the MSK SABR trial (NCT05534438) — trial-only, after systemic response

Likelihood of effect: Low for durable benefit off-protocol: randomized consolidation was negative in unselected patients (NRG-BR002 HR 1.31, PMID:33885704); trial-only, biology-selected.

Toxicity burden: Low (radiation modality; lesion-specific, no systemic AE profile)

Counter-productive MoA: Low (No mechanism blunts the systemic goal; the risk is null benefit from consolidation, per the negative randomized readouts.)

Overall: **The on-protocol route to the curative-intent framing the patient asked for, honestly tempered by two negative randomized consolidation trials — a trial-only layer on the systemic backbone, never a standalone.**

Key references: NCT05534438 · PMID 33885704 · PMID 34995128

Evidence in detail

SBRT to the solitary liver lesion as consolidation on the MSK SABR trial (NCT05534438) — trial-only, after systemic response

Chmura SJ et al. *JAMA Oncol* 2021

- **Disease context:** Newly oligometastatic breast cancer (1-4 metastases) of any subtype, on standard systemic therapy (1L+ consolidation) — Breast-specific cohort; 1-4 oligometastases; on standard systemic therapy; randomized to add consolidation vs continue systemic alone.
- **n:** 125
- **Toxicities:** any treatment-related AE (grade ≥ 3): 12.0% (n=65)
- **Efficacy:** HR_PFS 1.31; PFS_rate no difference at 3 yr

Khan SA et al. *J Clin Oncol* 2022

- **Disease context:**De novo stage IV breast cancer with intact primary tumor, no progression after 4-8 mo of optimal systemic therapy (1L+ consolidation) — Stage IV intact primary; 50% HR+/HER2-, 29% HER2+, 10% TNBC, 11% other; bone-only 31%, viscera-only 26%, both 27%; median age 55.
- **n:** 256
- **Toxicities:**surgical complications (grade any): 15.0% (n=125)
- **Efficacy:**HR_OS 1.11 (95% CI 0.82-1.52); **OS_rate** 68.4%; **OS_rate** 67.9%